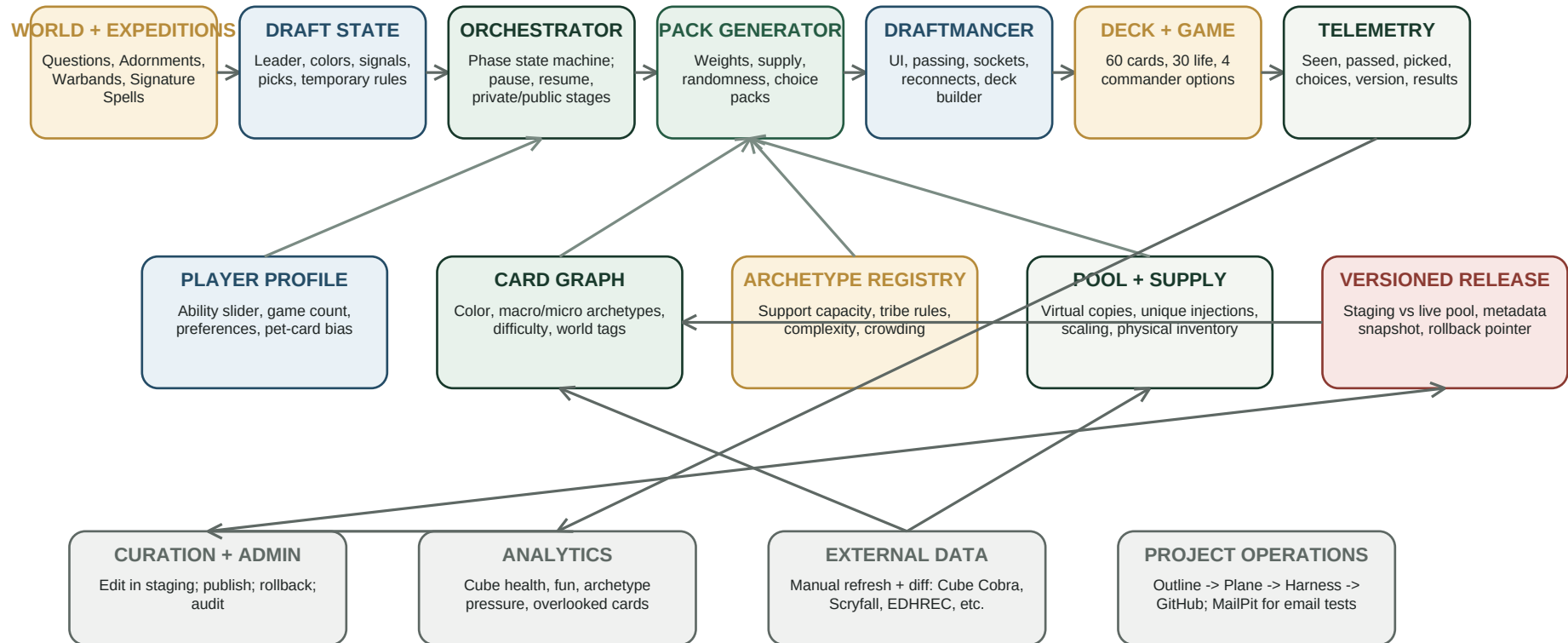


Gulchdale Systems Map v1.0

Working architecture and phased implementation plan



How to read this: Green decisions are established from the current design answers. Blue items are recommended V1 defaults. Amber items remain open and should become Plane tickets / Outline ADRs if they block implementation.

Architecture Contract

These boundaries are the most important part of the map. They prevent Gulchdale from becoming a hard-to-maintain Draftmancer fork or a collection of one-off scripts.

Established direction

- Gulchdale becomes the source of truth for its pool, metadata, rules, versions, player state, and telemetry.
- Cube Cobra remains useful as a public curation/publishing surface during migration.
- Draftmancer remains the multiplayer drafting/rendering foundation, not the owner of Gulchdale rules.
- The draft must support one- and two-color commanders, 60-card decks, four retained commanders, and player-count scaling.

Recommended architecture

- Gulchdale Draft Orchestrator sits above Draftmancer and owns the phase state machine.
- Pack Generator owns both passing packs and private Choice Packs.
- Choice Pack = private generated reward pool, pick N, destroy remainder, return to shared draft.
- Every live draft is tied to immutable Pool, Metadata, Rules, and Engine release identifiers.
- Staging is editable. Live is a pointer to a published release. Rollback moves the pointer.

Key unresolved design questions

- Exact commander game rule after four are retained (only marked leader in command zone vs another structure).
- Final launch archetype/tribe registry and capacity per archetype.
- Exact seven-ish stage draft recipe and number of total drafted nonland cards needed for satisfying 36-spell deckbuilding.
- Physical duplicate strategy for 6-8 player deployments.

1. World & Expeditions

Turns drafting decisions into an adventure. Owns locations, narrative questions, Expedition steps, Adornments, Warbands, Signature Spells, and player-visible consequences.

ESTABLISHED

- Choices should strongly influence archetypes and may influence tribal, aggro/control/midrange direction.
- Mechanical consequences should be discoverable within roughly 3-5 sessions: neither blatant nor invisible.
- Expeditions may permanently expand color identity for that draft through rewards such as Adornments.
- Current working concept uses more, smaller meaningful stages instead of four very large traditional packs; first instinct is about seven draft stages.
- World and location metadata is categorical and belongs in the Card Graph.

RECOMMENDED V1 DEFAULT

- Implement Expedition content as data-driven phase definitions, not hard-coded UI flows.
- Introduce a native Choice Pack event: a private 4-8 card generated pool shown to one player; keep the selected card/effect and destroy the remainder.
- Separate narrative text from mechanical effects so wording can change without changing game logic.
- V1 should include only 2-3 locations and one rule-changing reward (Adornment) to prove the system.

OPEN / NEEDS DECISION OR PLAYTEST

- Final number and pacing of intermissions after playtesting.
- Exact first Expedition script and which questions are shared versus player-specific.
- Whether some narrative choices can be mechanically neutral, or every choice must affect weights.
- How many permanent color-expansion effects can exist in one draft.

CORE DOMAIN OBJECTS

- ExpeditionDefinition, ExpeditionStep, Choice, ChoiceEffect, LocationTag, Adornment, Warband, SignatureSpell

2. Player Profile & Personalization

Stores long-term player tendencies without letting personalization overpower the current Expedition.

ESTABLISHED

- Accounts are not required for the first public version.
- Players explicitly select an ability level on a simple 1-10 slider.
- An internal, non-exposed metric tracks Gulchdale games played and can be compared with the self-rating.
- Pet-card bias should be mild; approximately 15% increased appearance probability is a reasonable ceiling to test.
- Questionnaire/Expedition signals must dominate personalization, preventing favorites from becoming a self-reinforcing rut.
- Gameplay decisions and outcomes are retained; real personal information is not shared.

RECOMMENDED V1 DEFAULT

- Create persistent anonymous player IDs first; add authenticated accounts later without changing draft data ownership.
- Store learned preferences as evidence-backed scores with confidence and sample count rather than a single permanent label.
- Allow a player to set an explicit favorite color/archetype later, but preserve learned preferences separately.
- Do not allow mid-draft difficulty changes.

OPEN / NEEDS DECISION OR PLAYTEST

- Exact retention period and deletion/export controls before a public release.
- Whether learned preferences are directly inspectable/editable by players.

CORE DOMAIN OBJECTS

- Player, PlayerIdentity, AbilitySelfRating, ExperienceMetric, PreferenceScore, PetCardAffinity, PlayerDraftHistory

3. Draft Session & Player Draft State

Owns everything true about a player in the current draft: commanders, leader, colors, current signals, temporary permissions, picks, and phase progress.

ESTABLISHED

- Players retain all four drafted commanders and mark one as the current Expedition leader.
- Primary color/archetype pivots happen through questionnaires/Expeditions.
- Gulchdale uses strictly one- and two-color commanders.
- Beginner-level players who choose a two-color leader should not normally receive an Adornment.
- Real-time deck-coherence scoring is not desired; recognizing coherence remains a drafting skill.
- Choosing a deliberately harder experience belongs in account/profile settings, not as an in-draft prompt.
- Winning, game count, and post-game feedback may inform the internal ability metric only between games.

RECOMMENDED V1 DEFAULT

- Model the draft as an explicit state machine rather than a fixed list of booster rounds.
- Snapshot all effective signals before each pack/choice generation so the admin can reconstruct why a card appeared.
- Distinguish permanent-for-draft permissions (Adornment) from temporary phase modifiers.

OPEN / NEEDS DECISION OR PLAYTEST

- Exact ability-level threshold that suppresses Adornments for two-color leaders.
- Whether any question can fully reset an archetype signal rather than merely add/subtract weight.

CORE DOMAIN OBJECTS

- DraftSession, DraftPlayerState, LeaderSelection, DraftSignal, TemporaryRule, ColorPermission, PhaseState

4. Card Pool & Supply

Defines which cards can exist in a draft, how many virtual copies are available, and which injections are personal/unique.

ESTABLISHED

- Gulchdale is singleton per final player deck, not singleton across the global draft pool.
- Virtual duplicate supply must scale with player count.
- Tribal archetypes receive targeted personal injections/boosters; the main pool contains support but not necessarily full critical mass.
- Two players may draft the same card into separate singleton decks.
- Commander-specific personal injections can be unique by design (for example Baba Lysaga land package).
- Physical inventory cost is a major constraint; \$500 would be exceptional and about \$1000 is a more plausible community target.

RECOMMENDED V1 DEFAULT

- Separate Virtual Supply from Physical Inventory. The engine should first prove gameplay; deployment reports later expose shortages/cost.
- Define copy policies: GLOBAL_UNIQUE, PER_PLAYER_SINGLETON, PLAYER_INJECTION_ONLY, UNLIMITED_EXCEPTION.
- Use player-count scaling rules per pool/archetype, not one global duplicate multiplier.
- Test 4-5 players first while keeping the data model capable of 2-8.

OPEN / NEEDS DECISION OR PLAYTEST

- How tribes remain fair above five players; likely allow each tribe to support two seats or expand the tribe roster.
- Exact pack scaling above four players and how often 16-card passing packs are required.
- Physical-copy reconciliation policy for public/store deployments.

CORE DOMAIN OBJECTS

- CardPoolEntry, SupplyPolicy, VirtualCopyCount, PhysicalInventory, PersonalInjection, PlayerCountScalingRule

5. Card Graph / Metadata

Describes what each card means to Gulchdale. This is the semantic backbone used by pack generation, curation, analytics, and world choices.

ESTABLISHED - V1 metadata: color; macro archetype (Aggro/Control/Midrange and Tribal flag); micro archetypes; primitive difficulty; world/location tags. - Human overrides outrank imported/derived values and are significant. - World/location tags are categorical. - Cards need negative as well as positive relationships (for example Blood Artist: Aristocrats high, Aggro low).	RECOMMENDED V1 DEFAULT - Use sparse normalized affinity values where 0.0 = actively anti-synergistic, 0.5 = neutral/default, 1.0 = strong affinity. Missing values resolve to 0.5; do not materialize every card x every dimension row. - Store raw imported source snapshots separately from derived Gulchdale scores. This preserves provenance, allows re-derivation, and prevents a source refresh from destroying human intent. - Store human overrides in a separate overlay table with optional reason, author, timestamp, and version. - Keep categorical tags many-to-many; do not force world tags into numeric affinity unless a location needs a weight.
OPEN / NEEDS DECISION OR PLAYTEST - Final controlled vocabulary for micro-archetypes and locations. - Whether card power becomes a required V1 metric or waits for external-data bootstrap.	
CORE DOMAIN OBJECTS - Card, CardPrinting, MetadataDimension, CardAffinity, CardTag, SourceSnapshot, DerivedScore, HumanOverride	

6. Archetypes, Difficulty & Deck Identity

Defines the intentionally supported ways to play Gulchdale, how many seats each can support, and how difficult each is to pilot.

ESTABLISHED - Supported archetypes are explicit, hard-coded design commitments maintained in a master registry. - General target: one drafter is strongly supported; two should be possible with visible scarcity signals; three should feel crowded. - Mono-color tribes receive dedicated commander/injection infrastructure and high main-pool weights. - Color count is the largest initial difficulty factor; Control is automatically more difficult; complex commanders can be hand-rated. - Modern do-everything cards should be heavily scrutinized; lower-cost/simpler cards may reduce archetype overlap. - Complex cards may remain when the play experience justifies them (Baba Lysaga is an example).	RECOMMENDED V1 DEFAULT - Give every archetype a configurable capacity profile: idealSeats, softCap, hardWarning, steeringStrength. - V1 overcrowding response: reduce further seeding and present a follow-up redirect question. Do not destroy already-drafted cards as an automatic punishment in V1. - Separate power from pilot difficulty. Difficulty can start with colors, archetype modifier, commander override, and later measured outcomes.
OPEN / NEEDS DECISION OR PLAYTEST - Per-archetype capacity values (Aggro may support more seats than Control). - Exact list of launch archetypes and tribes.	
CORE DOMAIN OBJECTS - Archetype, ArchetypeCapacity, Tribe, DifficultyProfile, CommanderComplexity, ArchetypeSupportPackage	

7. Pack Generator & Choice Generator

Creates public passing packs and private Choice Packs from current state, supply, weights, archetype capacity, and controlled randomness.

ESTABLISHED

- Generation is highly reactive; each decision may curate later pools and multiple pools must be tracked.
- Some stages may be entirely guided while others are intentionally unguided.
- Working overall target: about 35% of the finished draft experience should be entirely random; the remainder may be influenced to varying degrees.
- Full generation tree and weight explanation must be visible to admins but quiet to players.
- Over-seeding into scarce archetypes needs active handling.

RECOMMENDED V1 DEFAULT

- Use deterministic seeded randomness for reproducible debugging: same release + session seed + state produces the same candidate evaluation.
- Make guidanceStrength a per-phase setting rather than a global rule.
- Record candidate score components, not just final weight.
- Support two primitives: PassingPack and ChoicePack. ChoicePack is private, pick N, destroy remainder, never passes.
- For player-count scaling, configure pack size / discard threshold per phase rather than hard-coding a single 16/8 rule.

OPEN / NEEDS DECISION OR PLAYTEST

- Calibration of random/guided percentage after playtesting.
- Whether overload redirect questions target every player or only players entering the crowded archetype.

CORE DOMAIN OBJECTS

- PackDefinition, PassingPack, ChoicePack, CandidateScore, WeightContribution, GenerationTrace, RandomSeed, PoolReservation

8. Draft Orchestrator

Coordinates Gulchdale phases above Draftmancer: shared rounds, pauses, questions, private rewards, resumption, and completion.

ESTABLISHED - Gulchdale needs questionnaires and Expeditions between drafting stages. - Private pick-one reward pools are a desired core mechanic. - The draft should remain one coherent multiplayer session rather than repeatedly closing and rebuilding public sessions.	RECOMMENDED V1 DEFAULT - Add this as a first-class Gulchdale service. Do not bury orchestration inside the Draftmancer fork. - Use a state machine with phase kinds: PASSING_DRAFT, QUESTION, CHOICE_PACK, NARRATIVE, REVIEW, DECK_BUILD. - Draftmancer is commanded by the orchestrator and reports picks/events back; Gulchdale remains source of truth for phase state. - Allow admin pause/repair/resume for playtest recovery.
OPEN / NEEDS DECISION OR PLAYTEST - Exact behavior if one player disconnects during a private question/choice phase. - Whether all players must complete an intermission before any player can continue.	
CORE DOMAIN OBJECTS - DraftPlan, DraftPhase, PhaseTransition, OrchestrationState, PlayerBarrier, AdminRecoveryAction	

9. Draftmancer Adapter / Thin Fork

Preserves Draftmancer as the multiplayer drafting/rendering engine while exposing the minimum hooks Gulchdale needs.

ESTABLISHED

- Minimal maintenance and minimal dependence are primary goals.
- If upstream architecture eventually becomes incompatible, Gulchdale is willing to maintain the fork rather than abandon the experience.
- Generic upstream contributions are optional, not a requirement.

RECOMMENDED V1 DEFAULT

- Pin a known upstream commit and keep a clearly documented patch set.
- Draftmancer already supports per-booster picks/burns/discard rules, draft pause state, personal logs, and custom booster generation; reuse these before inventing replacements.
- Add only narrow hooks: pause/resume; inject/replace pending booster for a player; open/close Gulchdale intermission; inject selected private reward into drafted pool; emit complete pick/seen/pass telemetry.
- Keep Choice Pack rendering in the Gulchdale layer if possible; use Draftmancer card components, not its normal passing-pack lifecycle.
- Run upstream unit, manual statistical, and frontend test suites before accepting upstream merges.

OPEN / NEEDS DECISION OR PLAYTEST

- Exact files to patch after the harness performs a repository audit.
- Whether private Choice Pack UI can reuse client card components without coupling to DraftState.
- Long-term frequency of upstream merges.

CORE DOMAIN OBJECTS

- DraftmancerAdapter, UpstreamRevision, PatchManifest, CompatibilityTest, DraftmancerEventBridge

10. Deck Construction & Game

Defines the post-draft playable object and captures game outcomes without trying to simulate actual tabletop Magic.

ESTABLISHED - 30 starting life. - Final deck size is 60 cards: target 36 spells and 24 lands. - All four drafted commanders remain available after the draft; one is marked leader. - An Adornment communicates an off-color permission and currently exists physically as a draftable extra card; current tabletop practice can be preserved while digital rules become explicit. - Draftmancer mana-base recommendations may remain even though non-basic-land handling is limited. - Initial game results are entered by an admin/Game Master: winner, mana/color screw, archetype failures/successes/discoveries, fun/not-fun, and other observations.	RECOMMENDED V1 DEFAULT - Model Adornment permission explicitly in deck validation even if the physical Adornment remains a 61st reference card. - Add a lightweight post-game player questionnaire later: game too long, oppressive threat, MVP, deck fun. - Do not attempt sentence-level AI analysis in V1; store optional notes for later review.
OPEN / NEEDS DECISION OR PLAYTEST - Whether multiple commanders can be used in-command-zone during a game or only the marked leader (Charter currently defines availability, not exact game rule). - Exact legal role of the physical Adornment card in digital deck counts.	
CORE DOMAIN OBJECTS - Deck, DeckCard, CommanderOption, SelectedLeader, DeckPermission, GameRecord, PostGameFeedback	

11. Telemetry & Event Store

Creates an immutable history of what the system showed, what players did, and which version of Gulchdale produced the experience.

ESTABLISHED

- Cards seen but passed must be tracked, not only picks.
- Cube versions need major/minor concepts; major reconstruction may invalidate or flatten some learned metadata while minor changes can often continue.
- Post-game questions of interest include game length, oppressive threats, MVP, and deck fun.
- Low enjoyment and excessive game-length feedback attached to archetypes are important unhealthy-archetype signals.

RECOMMENDED V1 DEFAULT

- Mandatory events in V1: session created, release/version IDs, phase start/end, pack generated, candidates + weight trace, card seen, pick, pass, burn/discard, question shown/answered, reward granted, deck submitted, admin result entry.
- Use append-only event records plus derived read models; do not overwrite historical draft facts.
- Every draft stores immutable IDs/hashes for Pool Release, Metadata Release, Rules Release, and Engine Build. Historical analysis never reinterprets an old draft using today's metadata.
- Use semantic-ish releases: MAJOR = structural rules/pool reset; MINOR = compatible curation/content change; PATCH = metadata/text/config correction.

OPEN / NEEDS DECISION OR PLAYTEST

- Exact policy for carrying player preference evidence across a MAJOR release.
- Which events are retained indefinitely for a public deployment.

CORE DOMAIN OBJECTS

- TelemetryEvent, DraftEvent, GenerationTraceEvent, ReleaseSnapshot, GameFeedback, DerivedMetric

12. Analytics & Cube Health

Converts telemetry into signals for humans; it advises curation but does not automatically rebalance Gulchdale.

ESTABLISHED - Overlooked cards matter and should be discoverable from seen-vs-picked data. - Low fun, overlong games, and negative feedback associated with an archetype are health signals. - Analytics should support cube-health and player-tendency study.	RECOMMENDED V1 DEFAULT - V1 dashboard metrics: card seen/pick rate, pick position, final-deck inclusion, archetype occupancy, archetype fun, color screw, game length, oppressive-card flags, leader choice rate, choice-path frequency. - Never auto-remove cards from production from analytics alone. Create recommendations/flags for curator review. - Add confidence/sample-size display to avoid acting on three games as if they were thirty.
OPEN / NEEDS DECISION OR PLAYTEST - Formal thresholds for unhealthy cards/archetypes after enough playtest data exists. - How win rate should be weighted versus fun and political quality.	
CORE DOMAIN OBJECTS - CardHealthMetric, ArchetypeHealthMetric, PlayerTendencyMetric, HealthFlag, AnalyticsSnapshot	

13. Curation, Admin, Staging & Rollback

Becomes the long-term primary interface for maintaining Gulchdale: card list, metadata, archetypes, releases, imports, playtest notes, and rollback.

ESTABLISHED

- Live and staging/beta pools are required.
- Rollback should be deliberately easy for an admin but protected by strong confirmation.
- Plane should ideally have circular interaction with the app over time.
- Cube Cobra is currently the primary curation interface, so migration must be gradual.

RECOMMENDED V1 DEFAULT

- No direct edits to LIVE. All curation edits occur in STAGING; Publish creates an immutable release; rollback moves the live pointer to a prior release.
- All production-affecting changes are versioned. Reasons are required for high-impact overrides/removals, optional for routine metadata cleanup.
- V1 migration: import Cube Cobra as an upstream list source and detect when Cube Cobra is ahead. Later, invert the relationship so Gulchdale exports to Cube Cobra.
- Plane integration begins as deep links/ticket IDs stored on changes, not automatic ticket mutation.

OPEN / NEEDS DECISION OR PLAYTEST

- When Gulchdale becomes authoritative for card-list edits instead of Cube Cobra.
- Which change types require a second human review once delegation exists.

CORE DOMAIN OBJECTS

- StagingWorkspace, CurationChange, Release, Rollback, AuditEntry, PlaneReference, CubeCobraSyncStatus

14. External Data & Publishing Integrations

Imports external information intentionally and publishes selected Gulchdale views without making runtime gameplay depend on outside services.

ESTABLISHED

- External metadata is never silently refreshed in production.
- Refresh occurs on explicit admin command and/or release work; automatic checks may only flag that a source value changed.
- If a source disappears or changes schema, raise a flag; an admin may reconcile or intentionally protect/ignore the existing snapshot.
- Cube Cobra may need to inform V1 when its list is ahead.
- Game-store deployment is not a V1/V2 requirement.

RECOMMENDED V1 DEFAULT

- Every importer is snapshot -> parse -> validate -> diff -> admin review -> promote. Runtime drafts read only promoted Gulchdale data.
- Adapters should be replaceable and failures isolated by source.
- Publishing outputs should eventually include Cube Cobra export, Draftmancer-compatible export/debug file, card manifest, and build-cost report.

OPEN / NEEDS DECISION OR PLAYTEST

- Exact rights/terms policy per external data source before automated distribution.
- Final Cube Cobra write-back method and whether it remains one-way.

CORE DOMAIN OBJECTS

- ExternalSource, ImportSnapshot, ImportDiff, ImportFlag, PublishedArtifact, CubeCobraAdapter

15. Productization, Buildability & Portability

Makes Gulchdale reproducible outside the original playgroup without forcing those constraints into the core V1 game engine.

ESTABLISHED

- Community build target: approximately \$500 would be excellent; around \$1000 is considered plausible.
- Manabase is the major cost problem; spells are generally inexpensive.
- A future store might run Gulchdale locally or use a hosted server; dedicated computers/tablets may be preferable to phones.
- Game-store deployment is a stretch goal, not V1 or likely V2.

RECOMMENDED V1 DEFAULT

- Track cost/buildability metadata early, but defer store workflows.
- Generate shopping manifests from an immutable Pool Release so a physical build corresponds to software content.
- Eventually support a Community Build profile that swaps expensive printings/cards while preserving archetype role.

OPEN / NEEDS DECISION OR PLAYTEST

- What local-operator settings are safe to expose without breaking design assumptions.
- How many physical devices a store deployment should assume.

CORE DOMAIN OBJECTS

- BuildProfile, CardCostSnapshot, ReplacementCandidate, ShoppingManifest, DeploymentProfile

16. Project Operations & Toolchain

Connects architecture decisions to executable work in the new development environment.

ESTABLISHED - The development environment includes an AI Harness, Plane ticketing, Outline documentation, a GitHub repository, and MailPit. - AI assistance is expected across every project role where useful.	RECOMMENDED V1 DEFAULT - Outline is the canonical human-readable design/decision record: Charter, Systems Map, ADRs, rules, data dictionary, playtest reports. - Plane is the execution backlog: Epics map to systems/phases; tickets link to Outline decisions and GitHub branches/PRs. - GitHub is the canonical source repository and release history; protected main, feature branches, CI, tagged releases. - Harness executes Plane work against documented acceptance criteria; it must not invent game-design decisions when an OPEN item blocks implementation. - MailPit is local/test email capture only (account invites, resets, post-draft questionnaires); no production delivery dependency. - Every significant architecture change gets an ADR in Outline and a linked Plane ticket/PR.
OPEN / NEEDS DECISION OR PLAYTEST - Exact CI/CD target and staging host. - Whether Plane/Outline/GitHub connectors are automated in the first engineering phase or kept as disciplined links initially.	
CORE DOMAIN OBJECTS - ArchitectureDecisionRecord, PlaneEpic, PlaneTicket, GitHubPullRequest, BuildArtifact, TestEmail	

Draft Experience: Recommended V1 Shape

This is a development target, not a locked final draft recipe. Pack sizes and counts are balancing parameters. The goal is to prove the new rhythm before tagging or balancing the entire cube.

Stage	Purpose	Kind
0 - Lobby	Select release; 2-8 capable, test 4 first; player ability 1-10; anonymous identity allowed.	Shared
1 - Commander draft	Each player starts with 4 commanders; pick 1/pass until every player retains 4.	Passing
2 - Hire leader	Privately mark one of the four as Expedition leader; apply initial color/archetype signals.	Question
3 - Assemble crew	Creature-forward weighted drafting stage; commander synergy is strong but not deterministic.	Passing
4 - Bolster capabilities	Question: more crew / equipment / magic; updates weights.	Question
5 - Preparation / raid	One or more passing packs with player/leader signals; keep substantial randomness.	Passing
6 - Explore settlement	Choose Library / Forge / Graveyard / etc.	Question
7 - Reward	Private Choice Pack; may contain Adornment, Warband, Signature Spell, or location-themed card. Pick one, destroy rest.	Choice Pack
8 - Final drafting	Later passing stages react to all previous choices and current archetype capacity.	Passing
9 - Deck build	Build 60: target 36 spells + 24 lands; 30 life; four commanders retained, one leader marked.	Deck Build
10 - Game + feedback	Admin result entry + lightweight player questionnaire; telemetry closes the loop.	Post-game

Calibration questions for playtest

- How many total nonland cards should a player see/draft to comfortably build 36 spells without returning to the old 80-card excess?
- Which phases should use passing and which should use private Choice Packs?
- Does the story cadence feel like an adventure or like interruptions?
- Can a player clearly perceive the consequence of choices within 3-5 sessions?
- Does a beginner get a powerful, pilotable deck without feeling railroaded?

Development Phases

The roadmap is dependency-driven and vertical. Each phase has an exit criterion so the Harness has a testable definition of done.

Phase 0 - Preserve, Audit, and Establish Guardrails

WORK

- Freeze/snapshot current Gulchdale exporter, current Draftmancer custom list, current landing/draft flow, and current Cube Cobra identifiers.
- Import the Design Charter and this Systems Map into Outline as canonical design docs.
- Create Plane epics, branch strategy, CI baseline, local dev bootstrap, and ADR template.
- Fork/pin Draftmancer at a known upstream commit; do not modify it yet.
- Capture a legacy end-to-end smoke test so the overhaul can always prove what it replaced.

EXIT CRITERION

- Done when a new developer/Harness can clone, boot, run tests, and reproduce the legacy draft without hidden manual steps.

PRIMARY RISK / CONTROL

- Risk: accidentally rewriting a working legacy flow before preserving it.
- Control: baseline snapshot + smoke test + pinned Draftmancer revision.

Phase 1 - Gulchdale Data Foundation

WORK

- Create the Gulchdale database and release/version model.
- Import the current Cube Cobra pool into staging and preserve card IDs/printing references.
- Add Card Pool/Supply, Card Graph, Archetype Registry, World tags, commanders, personal injections, and human override structures.
- Implement explicit external-source snapshots and manual refresh/diff workflow.
- Build read-only admin views before edit workflows.

EXIT CRITERION

- Done when Gulchdale can represent the current cube plus metadata independently of Draftmancer/Cube Cobra.

Phase 2 - Pack Generator as a Standalone Simulator

WORK

- Implement PassingPack and ChoicePack generation with seeded randomness.
- Implement weighted score composition, neutral/anti-synergy behavior, player-count scaling, supply reservation, and admin explain traces.
- Implement archetype capacity and V1 crowding behavior (reduce seed + redirect question flag).
- Build simulation commands/tests that generate thousands of packs without a browser.
- Start with a small curated test slice before tagging the full pool.

EXIT CRITERION

- Done when admins can explain exactly why every candidate entered a generated pack and replay it from the seed.

Phase 3 - Draft Orchestrator and Narrative Engine

WORK
- Implement the phase state machine and data-driven Expedition definitions. - Support QUESTION, NARRATIVE, PASSING_DRAFT, CHOICE_PACK, REVIEW, and DECK_BUILD phases. - Implement leader marking among four retained commanders and temporary/permanent-for-draft modifiers. - Implement one location choice and one Adornment flow. - Add admin pause/repair/resume tools.
EXIT CRITERION
- Done when a simulated player can complete a multi-stage Expedition without Draftmancer.
PRIMARY RISK / CONTROL
- Risk: orchestration logic leaks into Draftmancer. - Control: simulated orchestrator must work before Draftmancer integration.

Phase 4 - Thin Draftmancer Integration

WORK

- Audit Draftmancer Session/DraftState/client card UI and identify the minimum patch surface.
- Reuse existing picks/burns/discard/pause behavior for ordinary passing stages.
- Add the event bridge, dynamic pack handoff, intermission pause/resume, and private Choice Pack integration.
- Keep Gulchdale state authoritative; Draftmancer owns rendering/passing/network synchronization only.
- Run Draftmancer upstream unit/manual/frontend tests in CI; add Gulchdale compatibility tests.

EXIT CRITERION

- Done when a real 4-player browser session can pause for a Gulchdale question/choice and resume without losing draft state.

Phase 5 - First Playable Vertical Slice

WORK

- Target 4 players first (architecture remains 2-8 capable).
- Commander round: each player ultimately retains four commanders, then privately marks one leader.
- Run at least two meaningful questions and one private Choice Pack.
- Finish enough draft material to build a 60-card deck; retain mana-base helper; enforce current Gulchdale permissions.
- Record complete telemetry and allow admin post-game result entry.
- Playtest repeatedly before expanding content breadth.

EXIT CRITERION

- Done when this new flow is more fun/useful than the legacy four-pack draft for a real playgroup.

PRIMARY RISK / CONTROL

- Risk: expanding content before proving the new draft is fun.
- Control: repeatedly play the narrow vertical slice before broad metadata work.

Phase 6 - Curation UI, Staging, Publish, and Rollback

WORK

- Build staging edits for pool, metadata, archetypes, questions, and injections.
- Publish immutable releases; live drafts select a release; add the big red rollback action with confirmations.
- Implement Cube Cobra import/ahead detection as a transitional workflow.
- Attach Plane ticket/Outline ADR references to impactful changes.
- Add release diff viewer.

EXIT CRITERION

- Done when routine curation no longer requires editing source files and production can be rolled back safely.

Phase 7 - Telemetry, Cube Health, and Playtest Analytics

WORK

- Build card seen/passed/picked dashboards and final-deck inclusion tracking.
- Add archetype occupancy/crowding, fun score, game length, color screw, oppressive threat, and MVP views.
- Add sample-size/confidence warnings and admin flags rather than automatic rebalancing.
- Generate version-aware playtest reports.

EXIT CRITERION

- Done when curation meetings can be driven by actual draft/game evidence without losing qualitative judgment.

PRIMARY RISK / CONTROL

- Risk: overreacting to tiny datasets.
- Control: sample-size/confidence warnings and curator approval for all changes.

Phase 8 - Persistent Player Learning (Optional Accounts)

WORK

- Persist anonymous player identity across sessions where possible.
- Implement ability slider + internal experience metric.
- Learn preferences and pet-card affinity with evidence/confidence.
- Cap pet-card influence around the mild ~15% test target and keep Expedition signals dominant.
- Add authenticated accounts only when persistence, privacy, and recovery are worth the added scope.

EXIT CRITERION

- Done when repeat players feel recognizable to Gulchdale without the system railroading them.

Phase 9 - Advanced Gulchdale Systems

WORK
- Expand locations, Warbands, Signature Spells, tribe packages, color-opening rules, and player-specific questions. - Experiment with more sophisticated archetype crowding/territory conflicts only after V1 data exists. - Improve non-basic mana recommendations and difficulty estimation if playtests justify it. - Add multiple draft modes while keeping one shared Card Graph and telemetry model.
EXIT CRITERION
- Done when new game modes are data/config extensions rather than forks of core logic.

Phase 10 - Buildability and External Deployment

WORK
- Cost snapshots, replacement suggestions, shopping manifests, Community Build profile. - Evaluate 6-8 player physical supply requirements and store/device workflows. - Harden hosting, operator configuration, backups, privacy, and support docs. - Treat commercial/licensing strategy as a separate product/legal workstream.
EXIT CRITERION
- Done when another operator can reproduce a specific Gulchdale release without the original creator present.

Plane Epic Structure

Suggested initial top-level epics. Tickets should link to the relevant Outline section/ADR and to the GitHub branch/PR.

Epic	Name	Primary phase
GD-000	Architecture & Legacy Baseline	Phase 0
GD-100	Data Foundation & Releases	Phase 1
GD-200	Card Graph & External Data	Phases 1-2
GD-300	Pack Generator & Simulation	Phase 2
GD-400	Draft Orchestrator & Expeditions	Phase 3
GD-500	Draftmancer Adapter	Phase 4
GD-600	Admin / Curation / Publishing	Phase 6
GD-700	Telemetry & Analytics	Phases 5-7
GD-800	Player Personalization	Phase 8
GD-900	Deployment / Buildability	Phase 10

Ticket discipline

- Every ticket has acceptance criteria and names the system boundary it may change.
- If implementation hits an OPEN design question, create/resolve an ADR instead of inventing canon in code.
- One Plane ticket may map to multiple commits, but production-affecting behavior lands through reviewed PRs and tests.
- Playtest findings become Plane curation/design tickets, not direct hot-fixes to live data.

Harness Handoff: First Work Order

This is the recommended starting instruction after copying the conversation and artifacts into the Harness.

FIRST WORK ORDER

- 1. Read the Design Charter and Systems Map; do not implement yet.
- 2. Audit the existing Gulchdale repository and current Draftmancer integration/export scripts. Produce an architecture inventory.
- 3. Preserve the current system as a runnable legacy baseline and add a smoke test.
- 4. Pin/fork current upstream Draftmancer and document the exact revision.
- 5. Propose repository structure, local bootstrap, database choice/migrations, CI, and service boundaries matching this map.
- 6. Create Plane epics/tickets for Phase 0 and Phase 1 and an Outline ADR for any proposed deviation.
- 7. Implement Phase 0 only after the inventory is reviewed; do not begin full Card Graph tagging or deep Draftmancer modifications yet.

NON-NEGOTIABLE GUARDRAILS

- Gulchdale owns game logic; Draftmancer patches stay thin.
- No runtime dependency on live external metadata services.
- Random generation is reproducible and explainable to admins.
- No direct edits to live data; staging -> publish -> immutable release.
- Telemetry stores factual history append-only; analytics are derived.
- User experience choices should be discoverable but not mechanically blatant.
- Build the playable vertical slice before broadening scope.

DECISIONS THE HARNESS MUST NOT INVENT

- Exact final seven-stage draft recipe and pack counts.
- Exact launch archetype/tribe list and per-archetype capacity.
- Final multi-commander gameplay rule after drafting four commanders.
- Physical inventory policy for 6-8 players / game-store deployments.
- Final privacy/retention policy for public player accounts.